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COMMUNICATION

WiFinINA

 ARDUINO  LGPL-2.1 V1.9.1  Arduino  20/03/2025

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 <http://www.arduino.cc/en/Ref...>  info@arduino.cc

Enables network connection (local and Internet) with the Arduino MKR WiFi 1010, Arduino MKR VIDOR 4000, Arduino Uno WiFi Rev.2 and Nano 33 IoT.

With this library you can instantiate Servers, Clients and send/receive UDP packets through WiFi. The board can connect either to open or encrypted networks (WEP, WPA). The IP address can be assigned statically or through a DHCP. The library can also manage DNS.

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This library allows you to use the Arduino Uno WiFi Rev.2, Arduino Nano 33 IoT, Arduino MKR 1010 and Arduino MKR VIDOR 4000 WiFi capabilities. It can serve as either a server accepting incoming connections or a client making outgoing ones. The library supports WEP, WPA2 Personal and WPA2 Enterprise encryptions. This library supports all the same methods of the original WiFi library plus the connectSSL(). The WiFinINA library is very similar to the Ethernet and the WiFi library, and many of the function calls are the same.

To use this library:

```
1 #include <SPI.h>
2 #include <WiFinINA.h>
```

Examples

Several examples for the **WiFinINA** library are available from the [Examples from Libraries page](#).

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WiFi Class

Client Class

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Help

Description

Initializes the WiFiNINA library's network settings and provides the current status.

Syntax

```
1 WiFi.begin(ssid);  
2 WiFi.begin(ssid, pass);  
3 WiFi.begin(ssid, keyIndex, key)
```

Parameters

ssid: the SSID (Service Set Identifier) is the name of the WiFi network you want to connect to.

keyIndex: WEP encrypted networks can hold up to 4 different keys. This identifies which key you are going to use.

key: a hexadecimal string used as a security code for WEP encrypted networks.

pass: WPA encrypted networks use a password in the form of a string for security.

Returns

WL_CONNECTED when connected to a network

WL_IDLE_STATUS when not connected to a network, but powered on

Example

```
1 #include <WiFiNINA.h>  
2  
3 //SSID of your network  
4 char ssid[] = "yourNetwork";  
5 //password of your WPA Network  
6 char pass[] = "secretPassword"  
7  
8 void setup()  
9 {  
10  WiFi.begin(ssid, pass);  
11 }  
12  
13 void loop () {}
```

WiFi.end()

Description

Turns off the WiFi module. If `WiFi.begin()` was used to connect to an access point, the connection will be disconnected. If `WiFi.beginAP()` was used before to create an access point, the `WiFi.end()` will stop listening it too.

Syntax

```
1 WiFi.end();
```

Parameters

None

Returns

Nothing

WiFi.beginAP()

Description

Initializes the WiFiNINA library in Access Point (AP) mode. Other WiFi devices will be able to discover and connect to the created Access Point.

Syntax

```
1 WiFi.beginAP(ssid);  
2 WiFi.beginAP(ssid, channel);  
3 WiFi.beginAP(ssid, passphrase);  
4 WiFi.beginAP(ssid, passphrase,
```

Parameters

`ssid`: the SSID (Service Set Identifier) of the created Access Point. Must be 8 or more characters.

`passphrase`: optional, the WPA password of the created Access Point. Must be 8 or more characters.

channel 1;

Returns

WL_AP_LISTENING when creating access point succeeds

WL_CONNECT_FAILED when creating access point fails

Example

```

1  /*
2   WiFi Web Server LED Blink
3
4   A simple web server that le
5   This sketch will create a n
6   It will then launch a new s
7   to the Serial Monitor. From
8   to turn on and off the LED
9
10  If the IP address of your b
11   http://yourAddress/H turn
12   http://yourAddress/L turn
13
14   created 25 Nov 2012
15   by Tom Igoe
16   adapted to WiFi AP by Adafr
17  */
18
19  #include <SPI.h>
20  #include <WiFiNINA.h>
21  #include "arduino_secrets.h"
22  //please enter your sens
23  char ssid[] = SECRET_SSID;
24  char pass[] = SECRET_PASS;
25  int keyIndex = 0;
26
27  int led = LED_BUILTIN;
28  int status = WL_IDLE_STATUS;
29  WiFiServer server(80);

```

WiFi.beginEnterprise()

Description

Initializes the WiFiNINA library's network settings for a common WPA2 Enterprise network with username and password authentication (PEAP/MSCHAPv2).

Note: this feature requires NINA firmware version 1.3.0 or later. All string parameter supplied must have a combined length of under 4000 bytes.

Syntax

```
1 WiFi.beginEnterprise(ssid, user
2 WiFi.beginEnterprise(ssid, user
3 WiFi.beginEnterprise(ssid, user
```

Parameters

ssid: the SSID (Service Set Identifier) is the name of the WiFi network you want to connect to.

username: username part of WPA2 Enterprise (RADIUS) credentials

password: password part of WPA2 Enterprise (RADIUS) credentials

identity: WPA2 enterprise identity (optional)

ca: root certificate (string) to validate against (optional)

Returns

WL_CONNECTED when connected to a network

WL_IDLE_STATUS when not connected to a network, but powered on

Example

```
1  /*
2   This example connects to a
3   Then it prints the MAC addr
4   the IP address obtained, an
5
6   Based on ConnectWithWPA.ino
7  */
8  #include <SPI.h>
9  #include <WiFiNINA.h>
10
11 #include "arduino_secrets.h"
12 //please enter your sens
13 char ssid[] = SECRET_SSID; /
14 char user[] = SECRET_USER; /
15 char pass[] = SECRET_PASS; /
16 int status = WL_IDLE_STATUS;
17
18 void setup() {
19   //Initialize serial and wai
20   Serial.begin(9600);
21   while (!Serial) {
22     ; // wait for serial port
23   }
24
25   // check for the WiFi modul
26   if (WiFi.status() == WL_NO_
27     Serial.println("Communica
28   // don't continue
```

▼ **WiFi.disconnect()****Description**

Disconnects the WiFi from the current network.

Syntax

```
1 WiFi.disconnect()
```

Parameters

None

Returns

Nothing

▼ **WiFi.config()****Description**

WiFi.config() allows you to configure a static IP address as well as change the DNS, gateway, and subnet addresses on the WiFi shield.

Unlike WiFi.begin() which automatically configures the WiFi shield to use DHCP, WiFi.config() allows you to manually set the network address of the shield.

Calling WiFi.config() before WiFi.begin() forces begin() to configure the WiFi shield with the network addresses specified in config().

You can call WiFi.config() after WiFi.begin(), but the shield will initialize with begin() in the default DHCP mode. Once the config() method is called, it will change the network address as requested.

Syntax

```
1 WiFi.config(ip);
2 WiFi.config(ip, dns);
3 WiFi.config(ip, dns, gateway);
4 WiFi.config(ip, dns, gateway, s
```

Parameters

ip: the IP address of the device (array of 4 bytes)

dns: the address for a DNS server.

gateway: the IP address of the network gateway (array of 4 bytes). - optional: defaults to the device IP address with the last octet set to 1

subnet: the subnet mask of the network (array of 4 bytes). optional: defaults to 255.255.255.0

Returns

Nothing

Example

```
1 This example shows how to set
2
3 #include <SPI.h>
4 #include <Wi-FiNINA.h>
5
6 // the IP address for the shi
7 IPAddress ip(192, 168, 0, 177
8
9 char ssid[] = "yourNetwork";
10 char pass[] = "secretPassword
11
12 int status = WL_IDLE_STATUS;
13
14 void setup()
15 {
16     // Initialize serial and wa
17     Serial.begin(9600);
18     while (!Serial) {
19         ; // wait for serial port
20     }
21
22     WiFi.config(ip);
23
24     // attempt to connect to Wi
25     while ( status != WL_CONNEC
26         Serial.print("Attempting
27         Serial.println(ssid);
28     // Connect to WPA/WPA2 ne
29     status = WiFi.begin(ssid
```

Description

WiFi.setDNS() allows you to configure the DNS (Domain Name System) server.

Syntax

```
1 WiFi.setDNS(dns_server1)
2 WiFi.setDNS(dns_server1, dns_se
```

Parameters

dns_server1: the IP address of the primary DNS server

dns_server2: the IP address of the secondary DNS server

Returns

Nothing

Example

```
1 This example shows how to set
2
3 #include <SPI.h>
4 #include <WiFiNINA.h>
5
6 // the IP address for the shi
7 IPAddress dns(8, 8, 8, 8); /
8
9 char ssid[] = "yourNetwork";
10 char pass[] = "secretPassword
11
12 int status = WL_IDLE_STATUS;
13
14 void setup()
15 {
16     // Initialize serial and wa
17     Serial.begin(9600);
18     while (!Serial) {
19         ; // wait for serial port
20     }
21
22     // attempt to connect to Wi
23     while ( status != WL_CONNEC
24         Serial.print("Attempting
25         Serial.println(ssid);
26         // Connect to WPA/WPA2 ne
27         status = WiFi.begin(ssid,
28
29         // wait 10 seconds for --
```


WiFi.setHostname()**Description**

Sets the hostname of the module, the hostname is sent in WiFi.begin(...) when an IP address is requested from a DHCP server.

Syntax

```
1 WiFi.setHostname(hostname)
```

Parameters

hostname - new hostname to use

Returns

Nothing

Example

```
1  ...
2  WiFi.setHostname("MyArduino")
3
4  // attempt to connect to WiFi
5  while (status != WL_CONNECTED)
6  {
7    Serial.print("Attempting to connect to WPA/WPA2 network: ")
8    Serial.println(ssid);
9    // Connect to WPA/WPA2 network
10   status = WiFi.begin(ssid, password);
11
12   // wait 10 seconds for connection
13   delay(10000);
14 }
15 ...
```

▼ **WiFi.setTimeout()****Description**

Sets the connection timeout value in milliseconds for WiFi.begin(...).

Syntax

Parameters

timeout - the connection timeout value
in milliseconds

Returns

Nothing

Example

```
1  ...
2  WiFi.setTimeout(120 * 1000);
3
4  // attempt to connect to WiFi
5  while (status != WL_CONNECTED)
6  {
7      Serial.print("Attempting to connect to WPA/WPA2 network: ");
8      Serial.println(ssid);
9      // Connect to WPA/WPA2 network
10     status = WiFi.begin(ssid, password);
11
12     // wait 10 seconds for connection
13     delay(10000);
14 }
15 ...
```

▼ WiFi.SSID()

Description

Gets the SSID of the current network

Syntax

```
1 WiFi.SSID();  
2 WiFi.SSID(wifiAccessPoint)
```

Parameters

wifiAccessPoint: specifies from which network to get the information

Returns

A string containing the SSID the WiFi is currently connected to.

A string containing name of network requested.

Example

```
1 #include <SPI.h>  
2 #include <WiFiNINA.h>  
3  
4 //SSID of your network  
5 char ssid[] = "yourNetwork";  
6 int status = WL_IDLE_STATUS;  
7  
8 void setup()  
9 {  
10   // initialize serial:  
11   Serial.begin(9600);  
12  
13   // scan for existing network  
14   Serial.println("Scanning available networks");  
15   scanNetworks();  
16  
17   // attempt to connect using  
18   Serial.println("Attempting to connect to WPA SSID: ");  
19   status = WiFi.begin(ssid);  
20  
21   Serial.print("SSID: ");  
22   Serial.println(ssid);  
23  
24 }  
25  
26 void loop () {}  
27  
28 void scanNetworks() {  
29   // Scan for networks  
30   Serial.println("Scanning for networks");  
31   WiFi.scanNetworks();  
32   int n = WiFi.scanComplete();  
33   if (n == -1) {  
34     Serial.println("Scanning complete");  
35   }  
36   if (n > 0) {  
37     Serial.print("Found ");  
38     Serial.print(n);  
39     Serial.println(" networks");  
40     for (int i = 0; i < n; i++) {  
41       Serial.print(i + 1);  
42       Serial.print(": ");  
43       Serial.print(WiFi.SSID(i));  
44       Serial.print(" (");  
45       Serial.print(WiFi.RSSI(i));  
46       Serial.print(" dBm)");  
47       Serial.println();  
48     }  
49   }  
50 }
```

WiFi.BSSID()

Description

Gets the MAC address of the router you are connected to or the MAC address of a network that was scanned.

Syntax

```
1 WiFi.BSSID(bssid)
2 WiFi.BSSID(wifiAccessPoint, bss
```

Parameters

bssid - 6 byte array

wifiAccessPoint - specifies from which network to get the information (optional), only needed after a scan

Returns

A byte array containing the MAC address of the router the WiFi shield is currently connected to or the MAC address of a network that was scanned. The first array index contains the last byte of the MAC address.

Example

```
1 #include <Wi-FiNINA.h>
2
3 //SSID of your network
4 char ssid[] = "yourNetwork";
5 //password of your WPA Network
6 char pass[] = "secretPassword";
7
8 void setup()
9 {
10   WiFi.begin(ssid, pass);
11
12   if ( status != WL_CONNECTED)
13     Serial.println("Couldn't
14     while(true);
15   }
16   // if you are connected, print
17   else {
18     // print the MAC address of
19     byte bssid[6];
20     WiFi.BSSID(bssid);
21     Serial.print("BSSID: ");
22     Serial.print(bssid[5], HEX);
23     Serial.print(":");
24     Serial.print(bssid[4], HEX);
25     Serial.print(":");
26     Serial.print(bssid[3], HEX);
27     Serial.print(":");
28     Serial.print(bssid[2], HEX);
29     Serial.print(bssid[1], HEX);
30     Serial.print(bssid[0], HEX);
31     Serial.println();
32   }
33 }
```

WiFi.RSSI()

Description

Gets the signal strength of the connection to the router

Syntax

```
1 WiFi.RSSI();
2 WiFi.RSSI(wifiAccessPoint);
```

Parameters

wifiAccessPoint: specifies from which network to get the information

Returns

long : the current RSSI/Received Signal Strength in dBm

Example

```
1 #include <SPI.h>
2 #include <Wi-FiNINA.h>
3
4 //SSID of your network
5 char ssid[] = "yourNetwork";
6 //password of your WPA Network
7 char pass[] = "secretPassword"
8
9 void setup()
10 {
11   Wi-Fi.begin(ssid, pass);
12
13   if (Wi-Fi.status() != WL_CONNECTED)
14     Serial.println("Couldn't get a Wi-Fi connection");
15   while(true);
16 }
17 // if you are connected, print the received signal strength
18 else {
19   // print the received signal strength in dBm
20   long rssi = Wi-Fi.RSSI();
21   Serial.print("RSSI:");
22   Serial.println(rssi);
23 }
24 }
25
26 void loop () {}
```

Wi-Fi.channel()

Description

Gets the Wi-Fi channel of a network that was scanned.

Syntax

```
1 Wi-Fi.channel(wifiAccessPoint)
```

Parameters

wifiAccessPoint - specifies from which network to get the information

Returns

Wi-Fi channel of scanned network

Example

```
1  ...
2  // scan for nearby networks
3  Serial.println("** Scan Net
4  int numSsid = WiFi.scanNetw
5  if (numSsid == -1)
6  {
7      Serial.println("Couldn't
8      while (true);
9  }
10
11 // print the list of networ
12 Serial.print("number of ava
13 Serial.println(numSsid);
14
15 // print the network number
16 for (int thisNet = 0; thisN
17     Serial.print(thisNet + 1)
18     Serial.print(" ")
19     Serial.print("Signal: ");
20     Serial.print(WiFi.RSSI(th
21     Serial.print(" dBm");
22     Serial.print("\tChannel:
23     Serial.print(WiFi.channel
24     byte bssid[6];
25     Serial.print("\t\tBSSID:
26     printMacAddress(WiFi.BSSI
27     Serial.print("\tEncryptio
28     printEncryptionType(WiFi.
29     Serial.print("\t\tECTD: "
```

WiFi.encryptionType()

Description

Gets the encryption type of the current network

Syntax

```
1  WiFi.encryptionType();
2  WiFi.encryptionType(wifiAccessP
```

Parameters

wifiAccessPoint: specifies which network to get information from

Returns

byte : value represents the type of encryption

TKIP (WPA) = 2

WEP = 5

AUTO = 8

Example

```
1 #include <SPI.h>
2 #include <Wi-FiNINA.h>
3
4 //SSID of your network
5 char ssid[] = "yourNetwork";
6 //password of your WPA Network
7 char pass[] = "secretPassword"
8
9 void setup()
10 {
11   Wi-Fi.begin(ssid, pass);
12
13   if ( status != WL_CONNECTED)
14     Serial.println("Couldn't g
15     while(true);
16   }
17   // if you are connected, pri
18   else {
19     // print the encryption typ
20     byte encryption = Wi-Fi.ency
21     Serial.print("Encryption Typ
22     Serial.println(encryption,HE
23   }
24 }
25
26 void loop () {}
```

▼ **Wi-Fi.scanNetworks()**

Description

Scans for available WiFi networks and returns the discovered number

Syntax

```
1 WiFi.scanNetworks()
```

Parameters

None

Returns

byte : number of discovered networks

Example

```
1  /*
2   * This example prints the board name and scans for available WiFi networks.
3   * Every ten seconds, it scans for available networks and prints the number of networks found.
4   * It does not connect to any network, so it is safe to use on public networks.
5   *
6   * Circuit:
7   *   * Board with NINA module (Arduino Uno)
8   *
9   * created 13 July 2010
10  * by dlf (Metodo2 srl)
11  * modified 21 Junn 2012
12  * by Tom Igoe and Jaymes Dec
13  */
14
15
16
17 #include <SPI.h>
18 #include <WiFiNINA.h>
19
20 void setup() {
21   //Initialize serial and wait for port openings
22   Serial.begin(9600);
23   while (!Serial) {
24     ; // wait for serial port
25   }
26
27   // check for the WiFi module
28   if (WiFi.status() == WL_NO_MODULE) {
29     Serial.println("WiFi module not found");
30   }
31 }
```

Ping a remote device on the network.

Syntax

```
1 WiFi.ping(ip);  
2 WiFi.ping(ip, ttl);  
3 WiFi.ping(host);  
4 WiFi.ping(host, ttl);
```

Parameters

ip: the IP address to ping (array of 4 bytes)

host: the host to ping (string)

ttl: Time of live (optional, defaults to 128). Maximum number of routers the request can be forwarded to.

Returns

WL_PING_SUCCESS when the ping was successful

WL_PING_DEST_UNREACHABLE when the destination (IP or host is unreachable)

WL_PING_TIMEOUT when the ping times out

WL_PING_UNKNOWN_HOST when the host cannot be resolved via DNS

WL_PING_ERROR when an error occurs

Example

```
1  /*
2   This example connects to an
3   Then it prints the MAC addr
4   the IP address obtained, an
5   Then it continuously pings
6
7   Circuit:
8   * Board with NINA module (A
9
10  created 13 July 2010
11  by dlf (Metodo2 srl)
12  modified 09 June 2016
13  by Petar Georgiev
14  */
15  #include <SPI.h>
16  #include <WiFiNINA.h>
17
18  #include "arduino_secrets.h"
19  //please enter your sens
20  char ssid[] = SECRET_SSID;
21  char pass[] = SECRET_PASS;
22  int status = WL_IDLE_STATUS;
23
24  // Specify IP address or host
25  String hostName = "www.google
26  int pingResult;
27
28  void setup() {
29    // Initialize serial and we
```

WiFi.status()

Description

Return the connection status.

Syntax

```
1  WiFi.status()
```

Parameters

None

Returns

WL_CONNECTED: assigned when connected to a WiFi network;

WL_AP_CONNECTED : assigned when a device is connected in Access Point mode;

WL_AP_LISTENING : assigned when the listening for connections in Access Point mode;

WL_NO_MODULE: assigned when the communication with an integrated WiFi module fails;

WL_IDLE_STATUS: it is a temporary status assigned when WiFi.begin() is called and remains active until the number of attempts expires (resulting in WL_CONNECT_FAILED) or a connection is established (resulting in WL_CONNECTED);

WL_NO_SSID_AVAIL: assigned when no SSID are available;

WL_SCAN_COMPLETED: assigned when the scan networks is completed;

WL_CONNECT_FAILED: assigned when the connection fails for all the attempts;

WL_CONNECTION_LOST: assigned when the connection is lost;

WL_DISCONNECTED: assigned when disconnected from a network;

Example

```
1 #include <SPI.h>
2 #include <WiFiNINA.h>
3
4 char ssid[] = "yourNetwork";
5 char key[] = "D0D0DEADF00DABB";
6 int keyIndex = 0;
7 int status = WL_IDLE_STATUS;
8
9 void setup() {
10     //Initialize serial and wait
11     Serial.begin(9600);
12     while (!Serial) {
13         ; // wait for serial port
14     }
15
16     // attempt to connect to Wi
17     while ( status != WL_CONNEC
18         Serial.print("Attempting
19         Serial.println(ssid);
20         status = WiFi.begin(ssid,
21
22         // wait 10 seconds for co
23         delay(10000);
24     }
25
26     // once you are connected :
27     Serial.print("You're connec
28 }
```

▼ WiFi.macAddress()

Description

Gets the MAC Address of your WiFi NINA module

Syntax

```
1 WiFi.macAddress(mac)
```

Parameters

mac: a 6 byte array to hold the MAC address

Returns

byte array : 6 bytes representing the MAC address of your module

Example

```
1 #include <SPI.h>
2 #include <WiFiNINA.h>
3
4 char ssid[] = "yourNetwork";
5 int status = WL_IDLE_STATUS;
6
7 byte mac[6];
8
9
10 void setup()
11 {
12   Serial.begin(9600);
13
14   status = WiFi.begin(ssid);
15
16   if ( status != WL_CONNECTED)
17     Serial.println("Couldn't
18     while(true);
19   }
20   // if you are connected, pr
21   else {
22     WiFi.macAddress(mac);
23     Serial.print("MAC: ");
24     Serial.print(mac[5],HEX);
25     Serial.print(":");
26     Serial.print(mac[4],HEX);
27     Serial.print(":");
28     Serial.print(mac[3],HEX);
29     Serial.print(":");
30     Serial.print(mac[2],HEX);
31     Serial.print(":");
32     Serial.print(mac[1],HEX);
33     Serial.print(":");
34     Serial.print(mac[0],HEX);
35     Serial.println();
36   }
37 }
```

Description

Returns the firmware version running on the module as a string.

Syntax

```
1 WiFi.firmwareVersion()
```

Parameters

None

Returns

The firmware version running on the module as a string

Example

```
1 ...
2 String fv = WiFi.firmwareVersion()
3 if (fv < "1.0.0") {
4   Serial.println("Please upgrade")
5 }
6 ...
```

▼ WiFi.lowPowerMode()

Description

Enable low power mode. This is an automatically managed mode where the WiFi NINA Module reduces its power drain bringing the overall power consumption to 30 mA. Any incoming data is received and the device sends out regularly the beacon signal each 100 ms to keep the AP connection alive.

Syntax

```
1 WiFi.lowPowerMode()
```

Returns

None

▼ WiFi.noLowPowerMode()

Description

Disables the power saving modes enabled with `lowPowerMode()`. This is the default status of Power Mode.

Syntax

```
1 WiFi.noLowPowerMode()
```

Returns

None

▼ WiFi.reasonCode()

Description

Return The deauthentication reason code.

Syntax

```
1 WiFi.reasonCode()
```

Parameters

None

Returns

The deauthentication reason code

Example

```
1  ...
2
3  while (status != WL_CONNECTED)
4      Serial.print("Attempting to connect to WPA/WPA2 SSID: " + ssid + " ");
5      Serial.println(ssid);
6      // Connect to WPA/WPA2 SSID: ssid with password password
7      status = WiFi.begin(ssid, password);
8      if (status != WL_CONNECTED)
9      {
10         Serial.print("Attempting to connect failed, ");
11         Serial.println(status);
12
13     }
14     // wait 10 seconds for next attempt
15     delay(10000);
16 }
17
18 ...
```

WiFi.hostByName()

Description

Resolve the given hostname to an IP address

Syntax

```
1 WiFi.hostByName(hostname, result)
```

Parameters

hostname: Name to be resolved

result: IPAddress structure to store the returned IP address

Returns

1 if hostname was successfully converted to an IP address, else the error code

Example


```
1  ...
2
3  while (status != WL_CONNECTED)
4      Serial.print("Attempting to connect to SSID: ");
5      Serial.println(ssid);
6      status = WiFi.begin(ssid, password);
7      delay(10000);
8  }
9  Serial.println("Connected to WiFi");
10 printWifiStatus();
11
12 Serial.println("\nStarting connection to server");
13 IPAddress result;
14 int err = WiFi.hostByName(serverName, result);
15 if(err == 1){
16     Serial.print("IP address: ");
17     Serial.println(result);
18 } else {
19     Serial.print("Error connecting to server: ");
20     Serial.println(err);
21 }
22
23 ...
```

WiFi.localIP()

Description

Gets the WiFi's IP address

Syntax

```
1  WiFi.localIP()
```

Parameters

None

Returns

the IP address of the board

Example

```
1 #include <WiFiNINA.h>
2
3 char ssid[] = "yourNetwork";
4
5 int status = WL_IDLE_STATUS;
6
7 IPAddress ip;
8
9 void setup()
10 {
11   // initialize serial:
12   Serial.begin(9600);
13
14   WiFi.begin(ssid);
15
16   if ( status != WL_CONNECTED
17       Serial.println("Couldn't
18       while(true);
19   }
20   // if you are connected, pr
21   else {
22     //print the local IP address
23     ip = WiFi.localIP();
24     Serial.println(ip);
25
26   }
27 }
28
29 void loop() { }
```

WiFi.subnetMask()

Description

Gets the WiFi's subnet mask

Syntax

```
1 WiFi.subnet()
```

Parameters

None

Returns

the subnet mask of the board

Example

```
1 #include <Wi-FiNINA.h>
2 int status = WL_IDLE_STATUS;
3
4 //SSID of your network
5 char ssid[] = "yourNetwork";
6 //password of your WPA Network
7 char pass[] = "secretPassword";
8
9 IPAddress ip;
10 IPAddress subnet;
11 IPAddress gateway;
12
13 void setup()
14 {
15   WiFi.begin(ssid, pass);
16
17   if ( status != WL_CONNECTED
18       Serial.println("Couldn't
19       while(true);
20   }
21   // if you are connected, print
22   else {
23
24     // print your subnet mask
25     subnet = WiFi.subnetMask(
26     Serial.print("NETMASK: ")
27     Serial.println();
28
29   }
```

Wi-Fi.gatewayIP()

Description

Gets the Wi-Fi's gateway IP address.

Syntax

```
1 WiFi.gatewayIP()
```

Parameters

None

Returns

An array containing the board's gateway IP address

Example

```
1 #include <SPI.h>
2 #include <Wi-FiNINA.h>
3
4 int status = WL_IDLE_STATUS;
5
6 //SSID of your network
7 char ssid[] = "yourNetwork";
8 //password of your WPA Network
9 char pass[] = "secretPassword";
10
11 IPAddress gateway;
12
13 void setup()
14 {
15   Serial.begin(9600);
16
17   Wi-Fi.begin(ssid, pass);
18
19   if ( status != WL_CONNECTED )
20     Serial.println("Couldn't connect to network");
21   while(true);
22 }
23 // if you are connected, print the IP address
24 else {
25
26   // print your gateway address
27   gateway = Wi-Fi.gatewayIP();
28   Serial.print("GATEWAY: ");
29   Serial.println(gateway);
30 }
```

Wi-Fi.dnsIP()

Description

Returns the DNS server IP address for the device.

Syntax

```
1 Wi-Fi.dnsIP()
2 Wi-Fi.dnsIP(n)
```

Parameters

optional parameter n for the number of the DNS server to get the second DNS server

Returns

the DNS server IP address for the device (IPAddress).

Example

```
1 #include <WiFiNINA.h>
2
3 #include "arduino_secrets.h"
4 char ssid[] = SECRET_SSID;
5 char pass[] = SECRET_PASS;
6
7 IPAddress emptyIP;
8
9 void setup() {
10
11   Serial.begin(115200);
12   while (!Serial) {}
13
14   Serial.print("Attempting to
15   Serial.println(ssid);
16   int status = WiFi.begin(ssi
17   if ( status != WL_CONNECTED
18     Serial.println("Couldn't
19     while(true);
20   }
21
22   Serial.print("DHCP assigned
23   IPAddress dns1 = WiFi.dnsIP
24   if (dns1 == emptyIP) {
25     Serial.println("not set")
26   } else {
27     dns1.printTo(Serial);
28     Serial.println();
29     IPAddress dns2 = WiFi.dns2
```

WiFi.getTime()

Description

Get the time in seconds since January 1st, 1970. The time is retrieved from the WiFi module which periodically fetches the NTP time from an NTP server.

Syntax

```
1 WiFi.getTime();
```

Parameters

None

Returns

Returns the time in seconds since January 1st, 1970 on success. 0 on failure.

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